

Annotation Semantics of the ICSI MRDA Corpus

Dialogue-Act Labels, Flags
and Adjacency-Pair Links

Progress Report

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This report explains how we read the dialogue-act labels and the adjacency-pair field of the ICSI MRDA corpus, each decision taken and the evidence behind it, and what the resulting numbers show. Numbers marked with [†] come from exploratory checks run over the output; all others are produced by the annotation script.

Key findings

1. **The reading reproduces earlier counts.** On the speaker’s own side of a label, `cs` 3,041, `co` 790 and `cc` 436 match an independent `.dadb`-based analysis exactly; `aa` differs by 2.
2. **The manual’s colon rule gives the specified number of two-target acts.** Reading only the side of a colon that describes the whole utterance gives exactly 18; pooling both gives 22.
3. **Adjacency numbering restarts** 506 times, so a pair cannot be identified by its number. Closing open pairs at each restart cut implausibly long links from 326 to 9.
4. **Of 3,831 proposals, 1,589 have a resolved reply, 2,237 have no pair, 5 are malformed;** 95.7% of the replies lie within the next five rows.
5. **A few numbers do not reconcile** (1,455 earlier against 1,589 here, one extra act), and 154 responses are lost because they are ordered just before the act they answer.

1 Purpose and Approach

The project needs, for every dialogue act in the corpus, a class label, a way to tell proposals from other acts, and the acts that responded to each proposal. The raw annotation does not give these directly: a label is a compound string (general tag, specific tags, disruption forms, quoted material) and the adjacency field is a notation that has to be read in order. Each ambiguity was settled by evidence, in this order of authority: the labelling manual (Dhillon et al., 2004), the corpus paper (Shriberg et al., 2004), our earlier exploratory report (Project EDA report, 2026), which parsed a different source (the `.dadb` files) and is therefore an independent check, and finally measurement on the data. Where none of these decides, the choice is called a *convention*, its effect is measured, and it is flagged.

We start from a table with one row per dialogue act. The corpus holds 111,500 annotated elements in 75 meetings; a pipe bar separates acts that share one prosodic unit (Dhillon et al., 2004; Shriberg et al., 2004), so splitting at the bars gives 118,694 rows. Each row keeps its label, the adjacency string of its element, its text and its quality flags.

2 Reading the Label

A label has one general tag, optional specific tags joined by a caret (written in alphabetical order, p. 15), an optional disruption form after a period (pp. 16, 19–20), and, after a colon, quoted material (p. 21). The tag `z` stands alone (p. 17).

Only the speaker’s own side of a colon counts. The manual states that the act left of a colon describes the whole utterance and the act right of it only the quotation (p. 21); the earlier report adopted the same rule after finding that ignoring it credits quoted speech to the speaker. 1,512 rows (1.27%) contain a colon, and for 90 of them the quoted side would add a target class the speaker did not perform[†]. The specification states 18 acts with two target tags; the own side gives 18 and pooling both sides gives 22[†].

Seven classes, with partial accept and reject merged. Labels become `cs`, `co`, `aa`, `bk`, `ar`, `cc` or `other`, with `aap` counted as `aa` and `arp` as `ar`, as the specification requires. The manual supports this: its quick reference groups `aap` with `aa` as positive responses and `arp` with `ar` as negative ones (pp. 6–7), and a partial accept or reject differs from a full one only in agreeing or disagreeing with part of the earlier utterance (pp. 59, 62).

One label per act, with every target class kept. A row keeps all its target classes and receives one label: **other** if there are none, that class if one, and otherwise the highest of **cc**, **co**, **cs**, **ar**, **aa**, **bk**. Position in the label cannot decide, because tags are attached alphabetically (p. 15), and the manual does not rank classes, so the order is a *convention*. It rests on the manual’s grouping: **cs**, **co** and **cc** introduce a course of action, while **aa**, **ar** and **bk** answer something already said (pp. 6–7), and a commitment is an explicit promise of future action (p. 74). It affects 18 rows of 118,694 (**cs+ar** 7, **cs+aa** 6, **co+ar** 3, **aa+bk** 1, **aa+cc** 1); only the comparison between the two groups is exercised, and keeping the full set makes the choice reversible.

Disruption forms, nonspeech and unlabelled acts. The manual defines % as indecipherable speech, %- as interrupted, %- as abandoned, x as nonspeech and z as an act deliberately left unlabelled (pp. 17, 110–114). We store only %- and %- as disruption markers; % describes the audio, not an interruption, and, although the corpus paper groups x with % as indecipherable, the manual, being the reference, is followed. An act is non-speech only if its *whole* label is x: the manual says a disruption form is appended to an act that could still be labelled (pp. 19–20), and indeed 236 of 326 rows with an x are of the **s.x** kind and keep their act, nine of them proposals, leaving 90 non-speech rows[†]. The tag z gets its own flag, because it covers chatter, bleeped passages, digit reading and duplicate microphone channels (pp. 17, 113–114), which can contain speech: 2,422 of 2,442 such rows have transcribed words[†].

Empty acts are kept apart from parse errors. A row is empty if it has no text and no error-class quality flag; a row with an error flag is never marked empty, non-speech or unlabelled, so a parse failure cannot pass for data. This is an engineering decision rather than a manual one; no row currently has an error flag. In all, 324 rows are empty.

A proposal is any act with cs or co among its target classes, not merely a single label of **cs** or **co**, so eligibility does not depend on the convention above. All 3,831 proposals (3,041 **cs**, 790 **co**) coincide in this release.

3 Reading the Adjacency Field

An adjacency pair is two utterances by different speakers, a first part **a** and a second part **b** (p. 25). A string holds tokens separated by periods; a plus marks a continuation by the same speaker (p. 27), a hyphen and number distinguish speakers who share a part (p. 28), underscores mark a pair inserted later by the annotators (pp. 28–29), and backchannels are never given a pair (p. 30).

Tokens belong to the whole act, not to one part. The manual gives the single token 20b to the act **s^ar|s^nd** (p. 27). In the data, among 4,182 elements with both a pipe bar and an adjacency string, the number of tokens equals the number of parts in only 29.4% of cases[†], so assigning tokens to parts by position would usually be wrong.

Pairs and chains. Reading each meeting once in order, a plain **a** opens a pair; an **a** with a plus or a speaker number above one extends it; a **b** adds its act to a chain for its speaker number, a bare continuation extending the only chain. A pair is keyed by its number and its underscores. Each rule restates the manual (pp. 27–29).

Numbering restarts close all open pairs. The manual says numbers ascend and are never reused cyclically (p. 26), but the field reuses them, so a late response would attach to the wrong proposal. The corpus paper states that meetings were annotated in ten-minute chunks (Shriberg et al., 2004), which explains restarts. Numbering returns to 1 on 506 occasions in all 75 meetings; the first restart falls at a median of 606s[†]. When it does, we close all open pairs. This cut links to a response more than 100 rows away from 326 to 9, and those more than 600s away from 185 to 6[†]. The restarts are not strictly periodic (Section 5), so the rule rests on the manual’s ascending numbering and on this effect.

Each proposal gets one status. It is *malformed* if any piece of its string breaks the notation (111 of 54,220 pieces, 0.20%[†]), otherwise *ambiguous* if its pair cannot be resolved cleanly, *ok* if a response was found, and *no pair* in all other cases, so a parse failure is never reported as an absent reply.

4 Results

Table 1: Rows by label, and flags.

Label	Rows	Share	Flag	Rows
other	99,686	83.99%	Proposal	3,831
bk	7,282	6.14%	Two target classes	18
aa	6,339	5.34%	Non-speech (x)	90
cs	3,041	2.56%	Unlabelled (z)	2,442
ar	1,120	0.94%	Empty	324
co	790	0.67%	Changed by b/bk variant	15,177
cc	436	0.37%	Error-class flag	0

Table 2: Tag occurrences against the earlier analysis and the published distribution[†].

Tag	Own side		Both sides of a colon		
	Here	Earlier	Here	Earlier	Published
cs	3,041	3,041	3,051	3,051	3,046
co	790	790	830	830	829
aa	6,084	6,086	6,088	6,090	6,091
cc	436	436	440	440	433
ar	—	—	957	957	955
aap	—	—	262	262	270
arp	—	—	180	180	180

Of the 111,500 elements, 44,940 (40.3%) carry an adjacency string; its 54,220 pieces hold 24,469 opening and 29,640 responding tokens and 111 pieces that do not parse[†]. The walk builds 18,619 pairs, 394 of them without a response and 35 ambiguous (none belongs to a proposal).

Table 3: Outcome of the 3,831 proposal links.

Status	Proposals	Share
ok	1,589	41.5%
no pair	2,237	58.4%
no adjacency string	1,855	
only answers a pair	358	
pair without response	24	
malformed	5	0.13%
ambiguous	0	0%

Resolved proposals have one responding chain in 1,357 cases, two in 195, three in 31 and four in 6. A response lies within the next five rows for 1,521 of 1,589 (95.7%), a figure close to the 95% reported earlier (Project EDA report, 2026), and none lies further than 50 rows. The manual requires different speakers (p. 25): 1,581 resolved proposals have a response from another speaker

and 8 only from the proposer’s own speaker identifier[†], against 34 same-speaker pairs in 20,575 earlier.

5 Commentary on the Numbers

1. **Counts agree, with two small gaps.** There is one more act than before (118,694 against 118,693), and **aa** is 2 lower (6,084 against 6,086). The extra act sits in the total, since the 2,532 removed **z** and **x** acts match the earlier removal. One candidate is the single act whose type attribute is missing in the release[†]; the **aa** gap is not caused by the type normalisation[†]. Neither can be settled without the `.dadb` files.
2. **1,589 resolved proposals against 1,455.** We resolve 134 more (9.2%). Counting pairs instead of acts gives 1,443, near but not equal[†]. Our proposals with an opening role are 25 more than before (1,616 against 1,591). The earlier report discarded the original adjacency column because its numbers repeat within a meeting, which matches the restarts found here, so this release probably carries that older column; this is a hypothesis.
3. **58.4% of proposals have no pair**, mostly because the annotation gives none: 46.8% of **cs** and 54.8% of **co** acts have no adjacency string[†], against 48% and 55% earlier.
4. **431 responses find no open pair, and 154 precede their own opening act.** In 154 cases (36%) the matching opening token follows within ten rows[†], 109 times in the next row. Rows are ranked by start time, so a response that starts slightly early is read first and dropped. These are real links lost by the implementation, so 1,589 is a lower bound.
5. **Restarts are not periodic.** Of 431 gaps between restarts, 233 (54%) lie between 500 and 700s, as ten-minute chunks predict, but 171 (40%) are under 300s (Figure 1); in 159 of them the numbering had reached ten or more, as in meeting Bmr008, where restarts occur at 155, 224, 338, 408 and 498s[†]. The rule handles both patterns; the chunk explanation holds for the first restart and about half of the rest.
6. **Unlabelled and non-speech rows contain speech.** Of the 90 rows that are **x** alone, 74 have transcribed words[†], although the manual defines the tag as non-vocal noise; the flag follows the label, not the words. Of the 324 empty rows, 17 carry a response label (7 **aa**, 10 **bk**)[†].

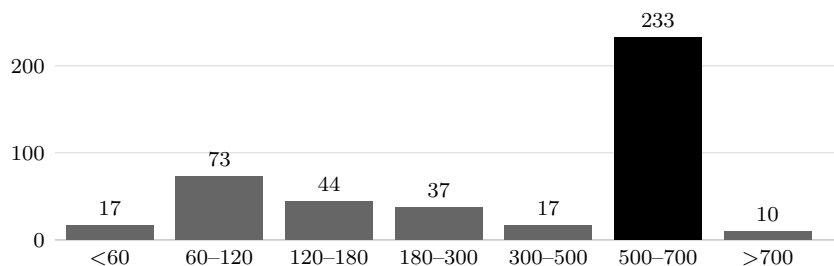


Figure 1: Seconds between consecutive numbering restarts in a meeting (431 gaps)[†]. The black bar is the range predicted by ten-minute chunks.

6 Limitations and Next Stage

Three points remain open for confirmation: the class order for multi-target acts, how non-speech, unlabelled and empty rows should count in evaluation and context building, and whether a trailing period in an adjacency string should be forgiven. Known weaknesses are the links lost to ordering

(item 4), undetected restarts to a number other than 1, and the unreconciled differences in item 2. The checks marked [†] are not yet scripted, no automated tests or type check have been run, and the provenance record names the earlier commit because the new code is not yet committed.

The coming stage will likely aim at exploiting these weaknesses and any others in the current implementation, so that no major problem surfaces later. We expect to test responses placed before their opening act, restarts to other numbers, cyclic numbering, malformed tokens and unusual label strings; to compare alternative restart rules and class orders on the same data; to join the output with the `.dadb` data to explain the extra act, the two `aa` acts and the 134 additional proposals; and to turn the exploratory checks into scripts.

References

- R. Dhillon, S. Bhagat, H. Carvey and E. Shriberg. *Meeting Recorder Project: Dialog Act Labeling Guide*. ICSI Technical Report TR-04-002, 2004.
- E. Shriberg, R. Dhillon, S. Bhagat, J. Ang and H. Carvey. The ICSI Meeting Recorder Dialog Act (MRDA) Corpus. In *Proc. 5th SIGdial Workshop on Discourse and Dialogue*, 2004.
- Structured Meeting Intelligence project. *Exploratory Analysis of the ICSI MRDA Corpus*. Progress report, September 2026.